The Hypergeometric Distribution

2026-04-17

Introduction

The hypergeometric distribution models the number of successes in a sample drawn without replacement from a finite population. It differs from the binomial by dependence across draws: removing a success reduces the probability of a subsequent success. It underlies Fisher's exact test and audit sampling.

Prerequisites

Combinatorics, binomial distribution.

Theory

A population of N items contains K "successes" and $N - K$ "failures". A sample of size n is drawn without replacement. Let X be the number of successes in the sample:

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \qquad \max(0, n - (N - K)) \le k \le \min(n, K).$$

Moments:

- $E[X] = nK/N$ (same as binomial with $p = K/N$).
- $\mathrm{Var}(X) = n \cdot \frac{K(N-K)}{N^2} \cdot \frac{N-n}{N-1}$.

The factor $(N - n)/(N - 1)$ is the **finite-population correction**; it shrinks the variance relative to binomial sampling, reflecting the reduced uncertainty when a substantial fraction of the population is sampled.

As $N \to \infty$ with $K/N = p$ fixed, hypergeometric converges to binomial: when the population is large relative to the sample, with/without replacement makes little difference.

Assumptions

- Finite population of known size N.
- Known number of successes K in the population.
- Simple random sampling without replacement.

R Implementation

```r
N <- 200; K <- 60; n <- 30

# PMF and moments
k <- 0:n
pmf <- dhyper(k, K, N - K, n)
c(mean = n * K / N,
  var  = n * K / N * (1 - K / N) * (N - n) / (N - 1),
  emp_mean = sum(k * pmf),
  emp_var  = sum((k - sum(k * pmf))^2 * pmf))
```

```r
# Compare with binomial: same mean, larger variance
c(binom_var = n * (K / N) * (1 - K / N))

# Fisher's exact test uses the hypergeometric
tab <- matrix(c(12, 18, 48, 122), 2, 2)
fisher.test(tab)
```

Output & Results

```
     mean       var   emp_mean    emp_var
      9.0     5.487       9.00      5.487

binom_var
     6.30

        Fisher's Exact Test for Count Data

data:  tab
p-value = 0.1412
alternative hypothesis: true odds ratio is not equal to 1
```

The hypergeometric variance is smaller than the binomial variance (5.49 vs 6.30) because the finite-population correction accounts for dependence. Fisher's exact test in the 2x2 example is a hypergeometric computation under the null of independence.

Interpretation

Hypergeometric inference appears whenever sampling without replacement matters. In quality audits, the exact probability of a given number of defective items being observed in a sample is hypergeometric. In contingency-table analysis of small cells, Fisher's exact test is hypergeometric.

Practical Tips

- For large populations ($N > 20n$), binomial approximation is typically adequate and simpler.
- For 2x2 contingency tables, Fisher's exact test is hypergeometric-based and preferable to chi-squared with small expected counts.
- `dhyper(x, m, n, k)` uses R's quirky naming: `m` = successes in population, `n` = failures in population, `k` = sample size.
- In survey sampling with stratification, the within-stratum variance uses a finite-population correction factor.
- Generalisations to multiple categories yield the multivariate hypergeometric; useful in categorical audit and card-game probabilities.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots